

de Sitter and Anti de Sitter Spacetime

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1 de Sitter Spacetime

de Sitter spacetime, dS_D , is the maximally symmetric solution of the vacuum Einstein equations with a positive cosmological constant $\Lambda > 0$. It has constant positive curvature and can be represented geometrically as a D -dimensional hyperboloid embedded in $(D + 1)$ -dimensional Minkowski spacetime,

$$-(X^0)^2 + \sum_{i=1}^D (X^i)^2 = \ell^2.$$

Its isometry group is $O(D, 1)$, with $D(D + 1)/2$ Killing vectors. Different coordinate systems cover different regions of the same de Sitter geometry and make different properties of the spacetime manifest.

Consider the $r = R$ case:

$$g_{tt} \rightarrow 0, \quad g_{rr} \rightarrow \infty.$$

In $\mathbb{R}^{D+1}$,

$$\sum_{i=1}^{D+1} (x^i)^2 = r^2 \quad \longrightarrow \quad D\text{-dimensional sphere.}$$

The symmetries of $\mathbb{R}^{D+1}$ consist of

$$D + 1 \text{ translations,} \quad \frac{D(D + 1)}{2} \text{ rotations.}$$

1.1 Embedding Space

Consider the $(D + 1)$ -dimensional Minkowski embedding space:

$$ds^2 = -(dX^0)^2 + \sum_{i=1}^D (dX^i)^2, \quad -(X^0)^2 + \sum_{i=1}^D (X^i)^2 = \ell^2 > 0.$$

The symmetry group is $O(D, 1)$. For comparison, a 2-dimensional sphere is

$$S^2 = \frac{O(3)}{O(2)}.$$

Solving the constraint for X^D ,

$$X^D = \pm \sqrt{\ell^2 + (X^0)^2 - \sum_{i=1}^{D-1} (X^i)^2},$$

where the quantity inside the square root is invariant under $O(D - 1, 1)$. The remaining symmetries are nonlinear:

$$\frac{D(D + 1)}{2} - \frac{D(D - 1)}{2} = D.$$

Thus,

$$\text{dS}_D = \frac{O(D, 1)}{O(D - 1, 1)}, \quad \dim(\text{dS}_D) = D.$$

It solves

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 0.$$

1.2 Global Coordinates

Take

$$X^0 = \ell \sinh t, \quad X^i = \ell \hat{n}^i \cosh t,$$

with

$$\sum_i \hat{n}^i \hat{n}^i = 1, \quad \sum_i \hat{n}^i d\hat{n}^i = 0, \quad \hat{n}^i \in S^{D-1}.$$

Then

$$ds^2 = \ell^2 [-dt^2 + \cosh^2 t d\Omega_{D-1}^2], \quad d\Omega_{D-1}^2 = d\hat{n}^i d\hat{n}^i.$$

This is in FLRW form with

$$a(t) = \cosh t, \quad \text{homogeneous and isotropic.}$$

For $D = 3$,

$$ds^2 = \ell^2 [-dt^2 + \cosh^2 t (d\theta^2 + \sin^2 \theta d\phi^2)].$$

1.2.1 Curvature

The spacetime is curved, $R_{\mu\nu\rho\sigma} \neq 0$, with

$$R = \frac{D(D-1)}{\ell^2}.$$

For a maximally symmetric spacetime,

$$R_{\mu\nu\rho\sigma} = \frac{R}{D(D-1)} (g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}) = \frac{1}{\ell^2} (g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

Therefore,

$$R_{\mu\nu} = \frac{D-1}{\ell^2} g_{\mu\nu}.$$

Using

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 0,$$

gives

$$\left[\frac{D-1}{\ell^2} - \frac{D(D-1)}{2\ell^2} + \Lambda \right] g_{\mu\nu} = 0 \quad \Rightarrow \quad \boxed{\Lambda = \frac{(D-1)(D-2)}{2\ell^2}}.$$

The metric is time-dependent in global coordinates:

$$\frac{\partial}{\partial t} \quad \text{is not a Killing vector.}$$

1.3 Static Coordinates

Static coordinates describe a patch of de Sitter spacetime in which the metric is independent of the time coordinate. The static patch covers the region accessible to an observer at fixed r , bounded by the cosmological horizon at $r = \ell$. In this region, ∂_t is a timelike Killing vector, making the time-translation symmetry explicit. At $r = \ell$, ∂_t becomes null and the static-coordinate metric becomes singular; this is a coordinate singularity corresponding to the de Sitter cosmological horizon rather than a physical singularity.

Take

$$X^a = r \hat{n}^a, \quad a = 1, 2, \dots, D-1, \quad \hat{n}^a \hat{n}^a = 1,$$

where $\hat{n}^a$ parametrizes a $(D-2)$ -sphere. The remaining embedding coordinates are

$$X^0 = \sqrt{\ell^2 - r^2} \sinh t, \quad X^D = \sqrt{\ell^2 - r^2} \cosh t.$$

Indeed,

$$-(\ell^2 - r^2) \sinh^2 t + (\ell^2 - r^2) \cosh^2 t + r^2 = \ell^2, \quad r < \ell.$$

Since $X^D > 0$ and $|\sinh t| < \cosh t$, this patch satisfies

$$|X^0| < X^D.$$

The induced metric is

$$ds^2 = -(\ell^2 - r^2)dt^2 + \frac{r^2}{\ell^2 - r^2}dr^2 + dr^2 + r^2 d\Omega_{D-2}^2 = -(\ell^2 - r^2)dt^2 + \frac{\ell^2}{\ell^2 - r^2}dr^2 + r^2 d\Omega_{D-2}^2.$$

Thus,

$$\frac{\partial}{\partial t} \quad \text{is a timelike Killing vector for } r < \ell.$$

The static coordinates become singular at $r = \ell$, although the spacetime itself is well-defined beyond this coordinate singularity.

1.3.1 Eddington–Finkelstein Coordinates

Define

$$u = t - \frac{1}{2\ell} \ln \left| \frac{\ell + r}{\ell - r} \right|, \quad du = dt - \frac{dr}{\ell^2 - r^2}.$$

Then

$$ds^2 = -(\ell^2 - r^2)du^2 - 2\ell du dr + r^2 d\Omega_{D-2}^2.$$

These coordinates are well-defined for $r > 0$, including $r < \ell$, $r = \ell$, and $r > \ell$. Moreover,

$$\frac{\partial}{\partial u}$$

is a Killing vector, timelike for $r < \ell$, and

$$r = \ell$$

is a horizon.

2 Anti-de Sitter Spacetime

Anti-de Sitter spacetime, AdS_D , is the maximally symmetric solution of the vacuum Einstein equations with a negative cosmological constant $\Lambda < 0$. It has constant negative curvature and can be represented as a D -dimensional hyperboloid embedded in a $(D+1)$ -dimensional flat spacetime with two timelike directions. Its isometry group is $O(D-1, 2)$, with $D(D+1)/2$ Killing vectors. Spatial slices of global AdS are noncompact, while timelike geodesics are confined by the AdS geometry. The universal cover of AdS is usually considered to remove the closed timelike curves associated with the periodic global time coordinate.

Consider a $(D+1)$ -dimensional embedding space with two timelike directions,

$$ds^2 = -(dX^0)^2 - (dX^D)^2 + \sum_{i=1}^{D-1} (dX^i)^2,$$

with isometry group $O(D-1, 2)$. Anti-de Sitter spacetime is defined by the isometric embedding

$$-(X^0)^2 - (X^D)^2 + \sum_{i=1}^{D-1} (X^i)^2 = -\ell^2.$$

For a constant-time section, introduce polar coordinates in the (X^0, X^D) plane,

$$X^0 = R \cos t, \quad X^D = R \sin t,$$

so that

$$ds^2 = -R^2 dt^2 - dR^2 + \sum_{i=1}^{D-1} (dX^i)^2.$$

The embedding equation becomes

$$-R^2 + \sum_{i=1}^{D-1} (X^i)^2 = -\ell^2,$$

so a fixed-time surface is a spatial hyperboloid and is noncompact.

Since t is periodic (angular-valued), the original AdS spacetime contains closed timelike loops. We therefore usually pass to the universal cover,

$$-\infty < t < \infty.$$

The covering space has isometry group $O(D-1, 2)$. As a coset,

$$\text{AdS}_D = \frac{O(D-1, 2)}{O(D-1, 1)},$$

where $O(D-1, 1)$ is the Lorentz group of D -dimensional Minkowski spacetime.

2.1 Global Coordinates

Introduce

$$X^0 = \ell \cosh \rho \cos t, \quad X^D = \ell \cosh \rho \sin t, \quad X^i = \ell \hat{n}^i \sinh \rho,$$

where $i = 1, 2, \dots, D-1$ and $\hat{n}^i$ parametrizes S^{D-2} . The metric becomes

$$ds^2 = \ell^2 [-\cosh^2 \rho dt^2 + d\rho^2 + \sinh^2 \rho d\Omega_{D-2}^2].$$

Here

$$\rho > 0, \quad 0 \leq t < 2\pi,$$

where the periodicity of t is removed by passing to the universal cover. Thus the original topology is

$$S^1 \times \mathbb{R}^{D-1},$$

while the universal cover has topology

$$\mathbb{R}^D.$$

2.2 Static Coordinates

The vector

$$\frac{\partial}{\partial t}$$

is a Killing vector with norm

$$\left\| \frac{\partial}{\partial t} \right\|^2 = -\ell^2 \cosh^2 \rho.$$

The Ricci scalar and cosmological constant are

$$R = -\frac{D(D-1)}{\ell^2}, \quad \Lambda = -\frac{(D-1)(D-2)}{2\ell^2} < 0.$$

Define

$$\sigma = \ell \sinh \rho, \quad \tau = \ell t.$$

Then the metric can be written as

$$ds^2 = -f(\sigma)d\tau^2 + \frac{1}{f(\sigma)}d\sigma^2 + \sigma^2 d\Omega_{D-2}^2, \quad f(\sigma) = 1 + \frac{\sigma^2}{\ell^2}.$$

2.3 Planar / Poincaré Coordinates

Write the embedding equation as

$$\eta_{\mu\nu} X^\mu X^\nu + (X^{D-1})^2 - (X^D)^2 = -\ell^2, \quad \mu = 0, 1, \dots, D-2.$$

Introduce

$$X^\mu = \ell e^\rho x^\mu.$$

Then

$$(X^D)^2 - (X^{D-1})^2 = \ell^2 + \ell^2 e^{2\rho} x^2 = \ell^2 (1 + e^{2\rho} x^2),$$

where

$$x^2 = \eta_{\mu\nu} x^\mu x^\nu.$$

From

$$(X^D)^2 - (X^{D-1})^2 = \ell^2 (1 + e^{2\rho} x^2),$$

we can choose

$$X^D + X^{D-1} = \ell e^{-\rho} (1 + e^{2\rho} x^2), \quad X^D - X^{D-1} = \ell e^\rho.$$

Hence,

$$X^{D-1} = \frac{\ell}{2} [e^{-\rho} + e^\rho (x^2 - 1)], \quad X^D = \frac{\ell}{2} [e^{-\rho} + e^\rho (x^2 + 1)],$$

so that $X^D > X^{D-1}$. The metric becomes

$$ds^2 = \ell^2 [d\rho^2 + e^{2\rho} \eta_{\mu\nu} dx^\mu dx^\nu].$$

It is invariant under

$$\rho \rightarrow \rho - \log \alpha, \quad x^\mu \rightarrow \alpha x^\mu.$$

Define $r = e^\rho$. Then

$$ds^2 = \ell^2 \left(\frac{dr^2}{r^2} + r^2 dx^2 \right), \quad 0 < r < \infty.$$

Equivalently, defining $z = 1/r$,

$$ds^2 = \frac{\ell^2}{z^2} (dz^2 + dx^2).$$

3 Geodesics

3.1 de Sitter Spacetime

The geodesic equation is

$$\frac{d^2 x^\mu}{d\sigma^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\sigma} \frac{dx^\beta}{d\sigma} = 0.$$

For a worldline $x^\mu(\sigma)$, define

$$\dot{x}^\mu \equiv \frac{dx^\mu}{d\sigma}, \quad g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -\kappa, \quad \kappa = \begin{cases} 1, & \text{massive particle (timelike),} \\ 0, & \text{light (null).} \end{cases}$$

Using the static metric, the geodesic action is

$$S_{\text{gs}} = \int d\sigma \left[-f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) \right],$$

with Lagrangian

$$\mathcal{L} = -f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2).$$

By spherical symmetry, choose the equatorial plane

$$\theta = \frac{\pi}{2}, \quad \dot{\theta} = 0,$$

so that

$$\mathcal{L} = -f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2 \dot{\phi}^2.$$

Since $\mathcal{L}$ only contains $\dot{t}$, t is cyclic. The conserved energy is

$$E = f(r)^2 \dot{t} = -\frac{1}{2} \frac{\partial \mathcal{L}}{\partial \dot{t}}, \quad \dot{t} = \frac{E}{f(r)^2}.$$

The energy is generated by the Killing vector

$$\xi = \frac{\partial}{\partial t}, \quad E = -\xi_\mu u^\mu.$$

Similarly, the conserved angular momentum is

$$L = \frac{1}{2} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = r^2 \sin^2 \theta \dot{\phi}.$$

For a massive particle,

$$-f(r)^2 \dot{t}^2 + f(r)^{-2} \dot{r}^2 + r^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) = -1.$$

Using the conserved quantities gives

$$\dot{r}^2 + V_{\text{eff}}(r) = E^2, \quad V_{\text{eff}}(r) = f(r)^2 \left(1 + \frac{L^2}{r^2} \right).$$

For de Sitter spacetime,

$$f(r)^2 = 1 - \frac{r^2}{R^2},$$

so

$$V_{\text{eff}}(r) = \left(1 + \frac{L^2}{r^2} \right) \left(1 - \frac{r^2}{R^2} \right).$$

The potential pushes the particle toward larger r .

For vanishing angular momentum, $L = 0$, the effective potential is an inverted harmonic oscillator. A particle initially stationary at $r = 0$ follows a geodesic. With nonzero initial radial velocity,

$$r(\sigma) = R \sqrt{E^2 - 1} \sinh \left(\frac{\sigma}{R} \right).$$

When the particle reaches $r = R$,

$$1 = \sqrt{E^2 - 1} \sinh \left(\frac{\sigma_*}{R} \right),$$

and therefore

$$\sigma_* = R \operatorname{arcsinh} \left(\frac{1}{\sqrt{E^2 - 1}} \right),$$

which is finite.

Meanwhile,

$$\dot{t} = \frac{E}{1 - r^2/R^2},$$

so as $r \rightarrow R$,

$$\dot{t} \rightarrow \infty.$$

Thus the static coordinates become singular at the horizon.

For nonzero angular momentum, $L \neq 0$,

$$V_{\text{ds}}(r) = \left(1 + \frac{L^2}{r^2} \right) \left(1 - \frac{r^2}{R^2} \right) = 1 - \frac{L^2}{R^2} - \frac{r^2}{R^2} + \frac{L^2}{r^2}.$$

As $r \rightarrow 0$,

$$V_{\text{ds}} \rightarrow \infty,$$

so particles with $L \neq 0$ cannot reach $r = 0$. Moreover,

$$\frac{dV_{\text{ds}}}{dr} = -\frac{2r}{R^2} - \frac{2L^2}{r^3} < 0.$$

3.2 Anti-de Sitter Spacetime

For anti-de Sitter spacetime,

$$\dot{r}^2 + V_{\text{eff}}(r) = E^2, \quad V_{\text{eff}}(r) = \left(1 + \frac{L^2}{r^2}\right) \left(1 + \frac{r^2}{R^2}\right).$$

For $L = 0$,

$$V_{\text{AdS}}(r) = 1 + \frac{r^2}{R^2}, \quad \dot{r}^2 = E^2 - 1 - \frac{r^2}{R^2}.$$

Differentiating gives

$$\ddot{r} = -\frac{r}{R^2},$$

which is a harmonic oscillator. At the turning point $\dot{r} = 0$,

$$r_{\text{max}} = R\sqrt{E^2 - 1},$$

and the trajectory can be written as

$$r(\sigma) = R\sqrt{E^2 - 1} \sin\left(\frac{\sigma - \sigma_0}{R}\right).$$

Thus, a massive particle with finite energy cannot reach $r = \infty$.

For $L \neq 0$, the terms

$$\frac{L^2}{r^2} \quad \text{and} \quad \frac{r^2}{R^2}$$

prevent the particle from reaching $r = 0$ and $r = \infty$, respectively, giving a U-shaped potential well. The minimum satisfies

$$\frac{dV_{\text{AdS}}}{dr} = 0 \quad \implies \quad \frac{2r}{R^2} - \frac{2L^2}{r^3} = 0,$$

so

$$r^4 = R^2 L^2, \quad r_*^2 = R|L|.$$

This gives a circular geodesic with

$$V_{\text{min}} = \left(1 + \frac{|L|}{R}\right)^2.$$

3.2.1 Null Geodesics

For null geodesics, $\kappa = 0$, and the effective potential is

$$V_{\text{null}}(r) = \frac{L^2}{r^2} \left(1 + \frac{r^2}{R^2}\right) = \frac{L^2}{r^2} + \frac{L^2}{R^2}.$$

As $r \rightarrow \infty$,

$$V_{\text{null}} \rightarrow \frac{L^2}{R^2},$$

so the potential does not diverge at infinity. Thus, light can reach arbitrarily large r .

For $L = 0$, use

$$r = R \sinh \rho.$$

The radial null geodesic satisfies

$$R\dot{\rho} = \pm \frac{E}{\cosh \rho},$$

and hence

$$R \sinh \rho = E(\sigma - \sigma_0).$$

For the coordinate time, define

$$E = \cosh^2 \rho \dot{t},$$

so

$$\dot{t} = \frac{E}{\cosh^2 \rho} = \frac{E}{1 + \frac{E^2(\sigma - \sigma_0)^2}{R^2}}.$$

Integrating,

$$R \tan\left(\frac{t}{R}\right) = E(\sigma - \sigma_0).$$

Therefore,

$$\sigma \rightarrow \infty \quad \implies \quad t \rightarrow \frac{\pi R}{2}.$$

Thus a radial null geodesic reaches $r \rightarrow \infty$ only at infinite affine parameter, but at finite global coordinate time.